

THIRUVENKADAPRASATH.S

192312016

8 Write a program to implement Linear Regression (LR) algorithm in python

CSV FILE

YearsExperience,Salary

1.1,39343

1.3,46205

1.5,37731

2.0,43525

2.2,39891

2.9,56642

3.0,60150

3.2,54445

3.2,64445

3.7,57189

3.9,63218

4.0,55794

4.0,56957

4.1,57081

4.5,61111

4.9,67938

5.1,66029

5.3,83088

5.9,81363

6.0,93940

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class LinearRegressionScratch:
```

```
    def __init__(self, learning_rate=0.01, iterations=1000):
```

```
        self.lr = learning_rate
```

```
        self.iterations = iterations
```

```
        self.weight = None
```

```
        self.bias = None
```

```
    def fit(self, X, y):
```

```
        num_samples = len(X)
```

```
        self.weight = 0
```

```
        self.bias = 0
```

```
        for _ in range(self.iterations):
```

```
            y_predicted = self.weight * X + self.bias
```

```
            dw = (2 / num_samples) * np.sum(X * (y_predicted - y))
```

```
            db = (2 / num_samples) * np.sum(y_predicted - y)
```

```
            self.weight -= self.lr * dw
```

```
            self.bias -= self.lr * db
```

```
    def predict(self, X):
```

```
        return self.weight * X + self.bias
```

```
df = pd.read_csv("linear_regression_data.csv")
```

```
X = df["YearsExperience"].values
```

```
y = df["Salary"].values
```

```
model = LinearRegressionScratch(learning_rate=0.01, iterations=1000)
```

```
model.fit(X, y)
```

```
predictions = model.predict(X)
```

```
mse = np.mean((y - predictions) ** 2)
```

```
r2_score = 1 - (np.sum((y - predictions) ** 2) / np.sum((y - np.mean(y)) ** 2))
```

```
print(f"Learned Slope (Weight): {model.weight:.2f}")
```

```
print(f"Learned Intercept (Bias): {model.bias:.2f}")
```

```
print(f"Mean Squared Error: {mse:.2f}")
```

```
print(f"R2 Score: {r2_score:.4f}")
```

```
sample_experience = 5.0
```

```
pred_salary = model.predict(sample_experience)
```

```
print(f"\nPredicted Salary for {sample_experience} years of experience: ${pred_salary:.2f}")
```

OUTPUT

